

Regime Aware Head Pruning

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Abstract—Most attention-head pruning methods assign each head a single global importance score, while input-adaptive approaches are often evaluated against static baselines that also change the selection criterion, obscuring where improvements come from. We conduct a controlled study of regime-conditioned head selection for time-series Transformers by inducing trend/seasonal/residual regimes via STL and applying structured head masks to PatchTST across multiple datasets, forecast horizons, sparsity levels, and random seeds.

We find that (i) global importance rankings can be unreliable at practical sparsity, whereas joint greedy elimination yields masks that preserve accuracy and can be trained from initialization without loss; (ii) once the selection criterion is held fixed, regime-based routing provides little benefit for deployable routers, despite an oracle that indicates substantial per-window gains are possible; and (iii) part of this potential can be recovered without explicit selection by ensembling predictions from diverse specialist masks. To support follow-up work on routing and conditional sparsity, we release per-window loss traces for all evaluated configurations.

Index Terms—Time-series forecasting, Transformers, attention mechanisms, structured pruning, model compression, conditional computation

I. INTRODUCTION

Transformer forecasters such as PatchTST [1] and iTransformer [2] dominate long-term time-series benchmarks, yet much of their multi-head attention is redundant: a substantial fraction of heads can be removed with little loss [3, 4]. The standard recipe scores each head with one global importance value and prunes the lowest-scoring ones. Time series, however, are not homogeneous: windows alternate between trend-dominant, seasonal-dominant and noise-dominant *regimes*, and a head that is useful in one regime can be harmful in another. This suggests a regime-conditional policy — detect the regime of each input window and activate the head subset selected for that regime — an instance of input-conditional computation [5] applied to head sparsity.

Prior comparisons of dynamic against static pruning share a structural weakness: the dynamic method and its static baseline differ simultaneously in *how* masks are selected and in *whether* they adapt to the input, so the observed gains cannot be attributed to adaptivity. We conduct, to our knowledge, the first controlled study that separates these factors, on a grid of 4 datasets $\times$ 4 horizons $\times$ 5 seeds $\times$ 6 sparsity levels ($\sim 9,300$ masked evaluations), with per-window losses stored for every configuration so that all statistics are recomputable offline.

Our contributions:

- 1) **Fragility of the standard method.** Global importance ranking degrades sharply at realistic sparsity and is beaten by *random* pruning on minute-resolution data; greedy backward elimination (joint selection) stays within $\sim 1\%$ of the unpruned model at 25% sparsity and degrades gracefully to 75% (Sec. V-A).
- 2) **Controlled attribution.** Regime-routed selection beats the standard static baseline in up to 15/16 cells (sign test $p=5 \times 10^{-4}$), but a pooled greedy mask — same criterion, no regime conditioning — reverses the comparison: the criterion, not the routing, explains the gain (Sec. V-B).
- 3) **The routing headroom is real but aleatoric.** A per-window oracle over the regime masks beats even the *unpruned* model in 15/16 cells (up to -12%), yet twelve deployable routers — STL-based, learned, bandit, gated, representation-based — fail to capture it. Near-oracle in-sample fits, flat $10\times$ learning curves and failed zero-shift diagnostics locate the failure in the signal, not the router (Sec. V-C).
- 4) **Combination instead of selection.** Averaging the specialist masks’ predictions beats the best static mask on all four datasets and surpasses the unpruned model on two; a soft-routing α -sweep shows performance *increases* as trust in the regime label decreases. An ablation attributes the gain to mask diversity, for which regime partitioning is the strongest but not a necessary source (Sec. V-D).
- 5) **Practical results.** With the mask imposed from scratch during training, 25% head pruning is free (-5.4% parameters, -12% FLOPs); a 12-feature classifier replaces per-window STL detection at $45\times$ lower latency with no end-to-end accuracy loss; and the findings transfer with a precise boundary to iTransformer’s cross-variate attention (Sec. V-F).

II. RELATED WORK

a) *Attention-head redundancy and static pruning.*:

The observation that multi-head attention is often over-parameterized dates to Michel et al. [3], who scored each head by the change in loss when it is masked and showed that many heads in NMT and BERT models can be removed with

little degradation [4]. Voita et al. [4] reached a similar conclusion using L_0 -regularized stochastic gates, and further argued that the remaining heads play specialized, interpretable roles. Subsequent work refined this picture in two ways that our study builds on. Li et al. [6] recast head pruning as a *subset-selection* problem and showed that greedy, per-head scoring (as in [3]) can underestimate how many heads may be removed because it ignores interactions between heads. Budhraj et al. [7] further reported that the relationship between measured head importance and actual prunability can be weak: randomly chosen head subsets often match importance-guided ones. We extend both critiques to time-series forecasting and sharpen them with a controlled comparison: a single mask chosen by *joint* greedy elimination remains stable across sparsity levels, whereas independent importance rankings can be unstable—to the point that random pruning can outperform them on minute-resolution data.

b) Dynamic and input-conditional pruning.: A parallel line of work makes the pruning decision depend on the input, a special case of conditional computation [5]. Token-level examples include learnable context dropping in autoregressive LLMs [8] and dynamic token sparsification in vision Transformers [9]. Closer to our setting are methods that adapt *head* usage per instance: SmartTrim prunes tokens and heads per input in vision-language models [10], hybrid dynamic pruning detects and skips unimportant heads at run time in hardware [11], delta-based reuse exploits temporal stability of neighbouring inputs on edge devices [12], and SPRINT selects which heads to prune per query, showing that targeted head removal can even *improve* task performance [13]. These works report gains from input-conditional sparsity over static baselines, but they typically do not control for the selection criterion: the dynamic method and the static baseline often differ both in *how* masks are chosen and in *whether* they adapt. Our study separates these factors and finds that, for regime-conditioned head selection in forecasting Transformers, the selection criterion accounts for most of the gain. At the same time, an oracle over the same per-regime masks indicates that input-conditional selection can retain substantial room for improvement, which none of the five practical routers we evaluate recovers.

c) Transformers for long-term time-series forecasting.: Our base model is PatchTST [1], which applies channel-independent patch attention and is a strong, widely used LTSF baseline; iTransformer [2] instead attends across variates, and linear models remain competitive on standard benchmarks [14], which cautions against purely architectural claims and motivates our efficiency-oriented framing. Structure-aware decomposition has a long history in classical time-series analysis: STL [15] separates trend, seasonal, and remainder components, and regime-switching models date back to [16]. Within deep forecasting, adaptivity to temporal structure also appears as architecture-level routing, e.g., multi-scale adaptive pathways in Pathformer [17]. We connect these threads at the level of *structural sparsity*: STL regimes define the conditioning signal, and head masks—not the architecture—provide the

adaptive mechanism.

d) Pruning and retraining.: Finally, our finding that training with the mask applied from initialization recovers (and on ETTm1 exceeds) the unpruned baseline, while post-hoc fine-tuning of a converged model fails, echoes the lottery-ticket literature [18]: the value of a sparsity pattern depends on when in training it is imposed. We did not search for winning initializations; we only show that, at 25% head sparsity, the pattern found on a converged model is trainable from scratch with no loss in accuracy.

III. MOTIVATION: REGIME-DEPENDENT HEAD STRUCTURE

Classical time-series analysis decomposes a series into trend, seasonal and remainder components [15], and regime-switching models formalize the observation that the data-generating process itself alternates between states [16]. If attention heads specialize — as they do in language models [4] — it is natural to ask whether their usefulness in a forecasting Transformer depends on which temporal regime the input window is in.

Three preliminary observations motivate the study (Fig. 1). First, within a trained PatchTST model, an average of 10.9 of 24 heads *flip the sign* of their leave-one-out importance across STL-labeled regimes: removing such a head hurts trend-dominant windows but helps residual-dominant ones, or vice versa. Second, this structure is causally exploitable: a mask selected for regime r achieves lower error on regime- r test windows than the other regimes’ masks in 12/16 dataset $\times$ horizon cells (Fig. 2), on weather in all 20/20 seed-cells. Third, the structure is model-specific: head importance *rankings* correlate strongly across regimes within one trained model ($\bar{r}_{\text{trend,seasonal}}=0.24$) but barely at all across training seeds ($\bar{r}\approx 0.05$) — which head plays which role is an accident of training, so any selection procedure must operate per model rather than transfer masks between models.

These observations make regime-conditional head selection plausible, but they do not establish that it *works*: a regime-routed method could gain from better selection rather than from routing, and per-window regime structure could be too noisy to exploit at inference time. The study is therefore organized around three questions:

- Q1 (criticism) How much of head-pruning quality is determined by the selection criterion alone — independent scoring vs. joint greedy elimination — across sparsity levels?
- Q2 (conditioning) With the criterion fixed, does conditioning the selection and its deployment on the detected regime add test-time accuracy?
- Q3 (headwork) How large is the gap between deployable regime-conditional methods and a per-window oracle over the same masks, and can any inference-time-legal mechanism close it?

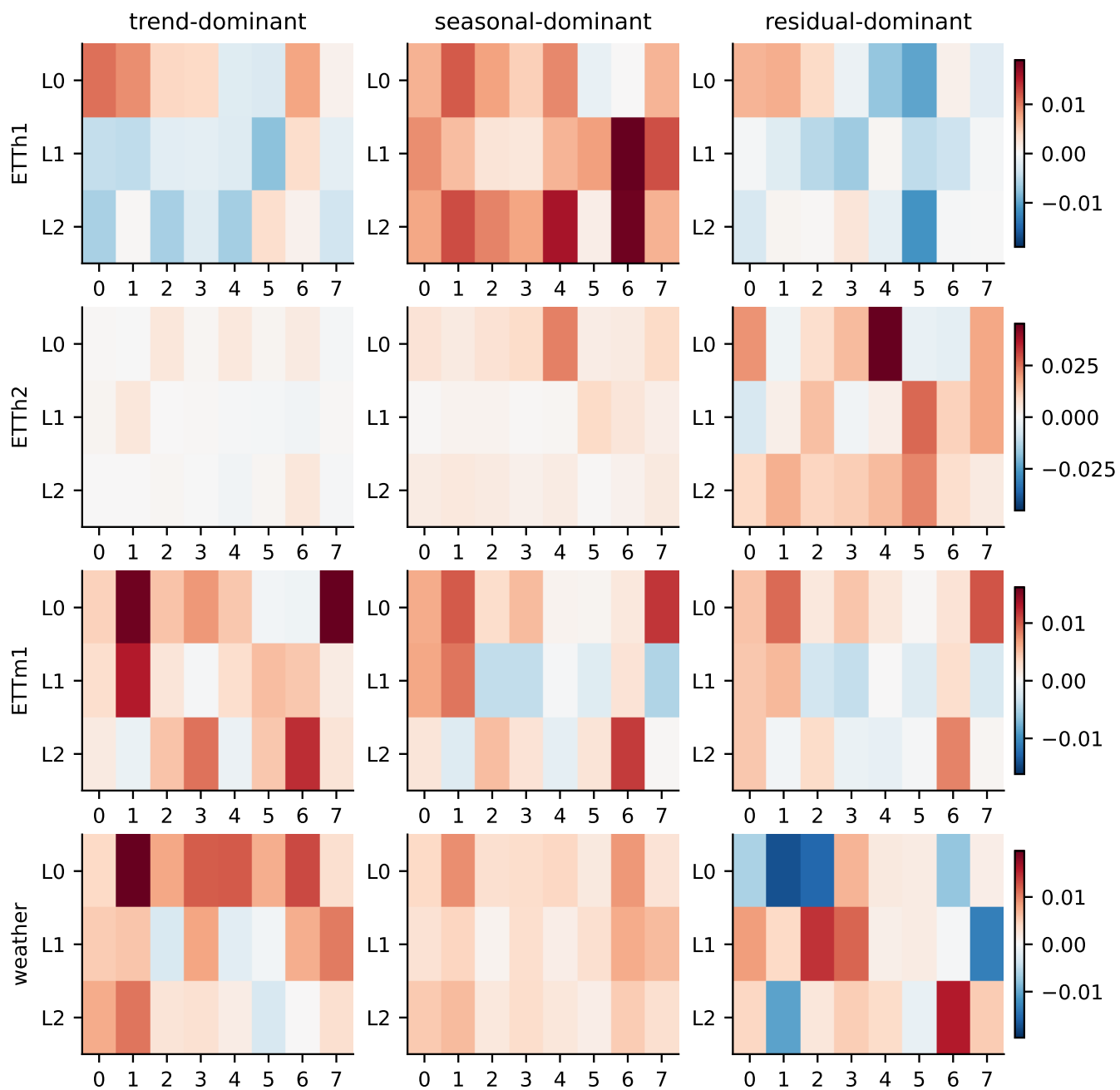


Fig. 1. Leave-one-out head importance (Δ validation MSE when the head is removed; red = removal hurts, blue = removal helps), averaged over 5 seeds, per STL regime (columns) and dataset (rows) at horizon 96. Column-to-column differences within a row are the regime effect; on average 10.9 of 24 heads flip sign across regimes.

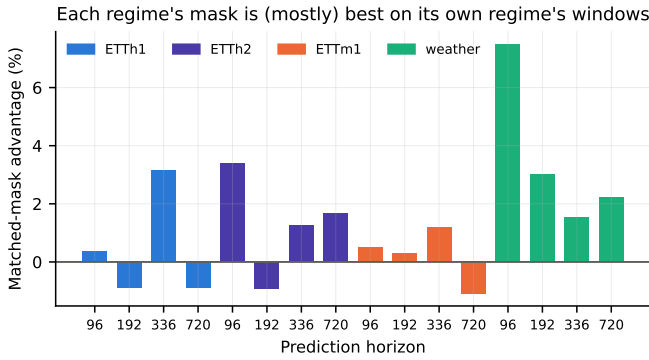


Fig. 2. Matched-mask advantage: relative error reduction of each regime’s specialist mask on its own regime’s test windows compared with the other regimes’ masks (25% sparsity, mean over 5 seeds). Positive in 12/16 cells; on Weather in all seed-cells.

IV. METHODS

A. Models, data and protocol

Our primary model is PatchTST [1] with $d_{\text{model}}=128$, $d_{\text{ff}}=256$, 3 encoder layers of 8 heads ($H=24$ heads total), input length 336, trained with the Time-Series-Library recipe (Adam, lr 10^{-4} , batch 32, 10 epochs, early stopping patience 3, lr halved per epoch). We evaluate on ETTh1, ETTh2, ETTm1 and Weather with prediction horizons $\{96, 192, 336, 720\}$ and 5 training seeds per cell (80 models). As a generalization test we repeat the core pipeline on iTransformer [2] at horizon 96 with the same capacity budget, noting that its heads attend across *variates* rather than time. All experiments run in a single pinned environment (library commit, dataset and checkpoint hashes recorded); validation and test loaders are unshuffled so window indices align with regime labels.

B. Regime labels

Each length-336 input window of the target variable is decomposed with robust STL [15] (period = one day at the dataset’s sampling rate). With component variances normalized to sum to one, the window’s regime is the argmax of $(s_{\text{trend}}, s_{\text{seasonal}}, s_{\text{residual}})$, and the *confidence margin* is the gap between the two largest scores. Labels are computed independently for train, validation and test splits; test regime mixes range from strongly trend-skewed (ETTh1: 93.6% trend) to strongly seasonal (Weather: $\geq 84\%$ seasonal).

C. Head masking and selection criteria

A head (ℓ, h) is deactivated by zeroing its attention output before the layer’s output projection; this functional masking is numerically equivalent to structurally removing the head’s parameter slices. All methods select masks on *validation* data only:

- **Independent importance (static).** Leave-one-out scoring [3]: importance of (ℓ, h) is the increase in validation MSE when it alone is masked; the lowest-scoring heads are pruned. Removal order = ascending importance.

- **Joint greedy (pooled).** Greedy backward elimination [6]: starting from all heads, repeatedly remove the head whose removal least harms validation MSE *given the removals so far*, on a fixed 256-window validation subsample.
- **Joint greedy (per-regime).** The same procedure restricted to validation windows of one regime, yielding three specialist masks; the dynamic policy applies the mask matching each window’s detected regime.
- **Baselines.** Random head subsets (3 draws per seed) and magnitude ranking by the ℓ_2 norm of each head’s projection slices.

Removal orders are recorded down to 6 remaining heads, so a mask at any sparsity in $\{12.5, 25, 37.5, 50, 62.5, 75\}\%$ is a prefix; all methods are compared at matched sparsity. Every evaluation stores per-window test and validation losses, enabling all subsequent analyses — routing, oracles, ensembling, statistics — without re-running models.

D. Routing, oracle and ensembles

At 25% sparsity the candidate set is $\mathcal{M} = \{\text{trend, seasonal, residual, pooled-greedy}\}$. The **oracle** routes each test window to its loss-minimizing mask (uses test targets; an upper bound, not a method). Deployable routers, all trained on validation only: (i) hard STL routing; (ii) a 12-feature gradient-boosted **regime detector** replacing STL at inference; (iii) **learned oracle-routers** — argmin classification, per-mask loss regression, and centered (regret) regression with margin weighting; (iv) **delayed-feedback** routing on observed regret of windows old enough for their targets to be available, and its sliding-window bandit variants; (v) **regret-gated** deviation from the pooled mask. Failure causes are separated by three diagnostics: in-sample fits (capacity), a learning curve over 2.2k–27k router-training windows using additionally computed *train-split* mask losses (data), and fit-on-half-of-test plus routing from mean-pooled encoder representations (shift / feature poverty).

Ensembles replace selection by combination: uniform averaging of the four masked predictions, the top-2 masks by validation MSE, weights $\propto$ inverse validation MSE, and *soft regime routing* in which the detected regime’s mask receives weight α (others share $1 - \alpha$), with $\alpha=1$ recovering hard routing. A **diversity ablation** rebuilds the pool with greedy masks from random validation subsamples (no regime information) under identical settings.

E. Retraining

To test whether pruning cost is intrinsic, the 25% masks are imposed *from initialization* and the model trained under the standard recipe (static mask; and a regime-conditional variant in which each training window applies its regime’s mask). This complements the earlier finding that fine-tuning an already-converged checkpoint under the mask fails.

F. Statistical analysis

Within each cell, methods are compared by seed-paired differences (Wilcoxon) and across the 16 cells by two-sided

sign tests. Cell-level uncertainty uses a hierarchical circular moving-block bootstrap (resampling seeds and 96-window blocks, 10^4 replicates, 95% CIs). Regime dependence of head importance is tested by circular rotations of the regime-label sequence, which preserve temporal autocorrelation. Reported oracle-gain *recovery* of a method m is $(L_{\text{pooled}} - L_m)/(L_{\text{pooled}} - L_{\text{oracle}})$.

V. RESULTS

A. The selection criterion dominates

Figure 3 shows the sparsity sweep at horizon 96; the full 16-cell grid is in Appendix Fig. 10. The standard independent-importance ranking is strikingly fragile: on ETTm1 it degrades test MSE by +15.4% already at 25% sparsity and by up to +46% at deeper cuts — worse than *random* pruning — and on ETTh1@720 it reaches +164%. Both greedy variants (pooled and per-regime) remain within $\sim 1\%$ of the unpruned baseline at 25% and degrade gracefully to 75% on every dataset. Magnitude ranking is consistently weak. Seed-level analysis explains the fragility: the six lowest-importance heads under independent scoring vary strongly across seeds, and ignoring head interactions makes the resulting mask high-variance — visible as the wide blue uncertainty bands in Fig. 3.

B. Attribution: criterion, not conditioning

Against the standard static baseline, regime-routed greedy selection wins 14/16 cells at 50% sparsity ($p=0.004$) and 15/16 at 62.5–75% ($p=5 \times 10^{-4}$) — exactly the kind of result reported in favour of dynamic pruning. The control reverses it: against the *pooled* greedy mask, dynamic wins only 0–4/16 cells at any sparsity (Fig. 4). The apparent advantage of the dynamic method is explained by its selection criterion; conditioning on the regime adds selection noise (minority regimes contribute few validation windows) without adding accuracy. The per-cell bootstrap (Fig. 11) confirms the picture at 25% sparsity: dynamic is significantly better than the standard static in 8/16 cells and never significantly worse, while static is significantly worse than the unpruned baseline in 11/16 cells.

C. The routing headroom is real but aleatoric

An oracle routing each test window to its best mask in $\mathcal{M}$ outperforms the *unpruned* model in 15/16 cells, by up to -12.3% (ETTh2@96: 0.2753 vs 0.3140) — a 25%-pruned network contains more usable capacity than the full model, if routed perfectly. No deployable router captures it (Fig. 5): hard STL routing, the learned detector, four learned oracle-routers, delayed-feedback and bandit variants, and regret-gating all recover between -33% and -3% of the oracle gain, i.e. none beats simply using the pooled mask ($\leq 6/16$ cells each).

Three diagnostics rule out fixable causes. *Capacity*: fitting the router in-sample reaches +96% recovery — the hypothesis class can express the routing function. *Data*: with train-split mask losses added, router training data grows $10\times$ (2.2k $\rightarrow$ 27k windows) while held-out regret stays flat (Fig. 6); the router memorizes its training windows (regret ≈ 0.002) at every scale.

Shift and features: fitting on the first half of the test set and predicting the second half fails equally, as does routing from the model’s own mean-pooled encoder representations (0/4 datasets). We conclude the per-window optimal-mask identity is aleatoric at inference time: it is decided by how a mask’s small perturbation interacts with the unpredictable component of the target.

D. Combination recovers what selection cannot

Averaging the specialist masks’ *predictions* sidesteps the prediction problem. The top-2 ensemble (two best masks by validation MSE; two pruned forward passes) beats the pooled mask on 4/4 datasets; inverse-MSE weighting recovers +8.7% of the oracle gain, ties the unpruned baseline on ETTm1 (0.2923 vs 0.2920) and beats it on ETTh2 (0.3070 vs 0.3140). Soft regime routing interpolates between hard routing and flat averaging: recovery falls monotonically as trust in the regime label rises — +9.2% at $\alpha=0.4$, +7.4% at 0.6, +1.8% at 0.8, negative at $\alpha=1$ (Fig. 7) — and regime-conditional weights merely tie the regime-blind ensemble. The diversity ablation (Fig. 12) completes the attribution: pools of greedy masks from *random* validation subsamples ensemble comparably on average (2/4 datasets each way; on ETTm1 the random pool also beats the unpruned baseline, 0.2909 vs 0.2920), but regime pools yield systematically deeper oracles (4/4 datasets). Regime information thus contributes *offline*, as the strongest single source of specialist diversity, while the extra oracle depth it creates is precisely the aleatoric part that remains unexploitable.

E. Pruning is free with retraining

Imposing the 25% mask from initialization erases the zero-shot cost entirely (Fig. 8): scratch-trained static-masked models reach 0.3770 vs 0.3745 unpruned on ETTh1@96 and 0.2905 vs 0.2920 (better) on ETTm1@96 — removing -5.4% of parameters and -12% of FLOPs at no accuracy cost, where zero-shot static pruning had cost up to +15%. The regime-conditional training variant does not improve on plain masked training (0.2977 vs 0.2905; a tested negative). This mirrors lottery-ticket observations [18]: the sparsity pattern’s value depends on when it is imposed.

F. Generalization to cross-variate attention

On iTransformer at horizon 96 (Fig. 9), head redundancy is far larger: on the 7-variate ETT datasets even 75% pruning costs only 0.1–2.6%, and on ETTh2 pruning *improves* the model at most ratios; only Weather (21 variates) shows PatchTST-like degradation. The criterion finding replicates (independent ranking again weakest), the oracle again beats the unpruned model on 4/4 datasets, and the ensembles again beat the pooled mask. One finding differs: regime-routed selection beats the pooled control in 4/4 datasets at mid sparsities — small margins on ETT, clearer on Weather (+1.9% vs +2.6% at 25% pruning) — so a modest regime-routing benefit survives the control on cross-variate attention, plausibly because regimes alter the between-variable correlation structure these

PatchTST, horizon 96

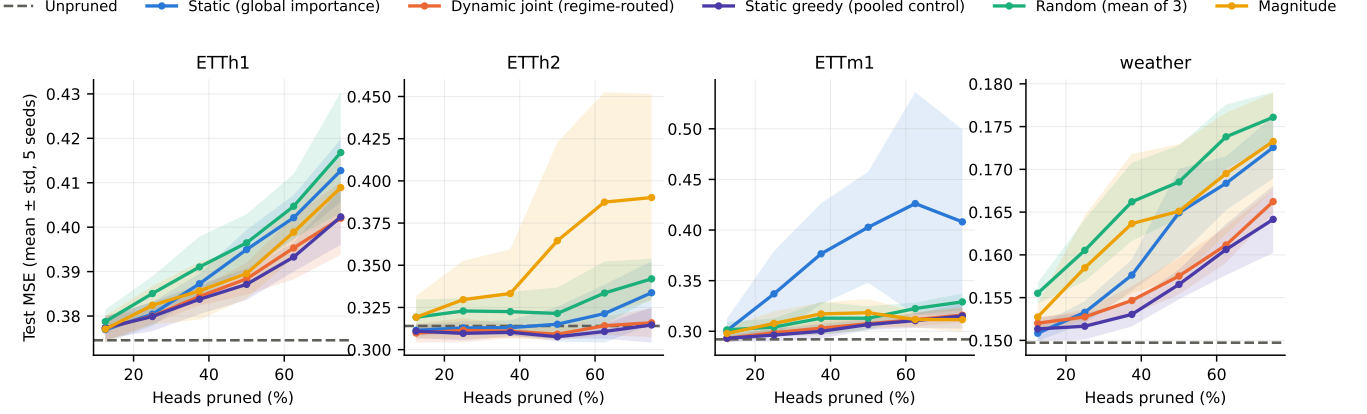


Fig. 3. Test MSE versus fraction of heads pruned (PatchTST, horizon 96; mean $\pm$ std over 5 seeds). The standard independent-importance ranking (blue) collapses on ETTm1 — worse than random — while both greedy variants (orange: per-regime; violet: pooled control) track the unpruned baseline (dashed) and degrade gracefully.

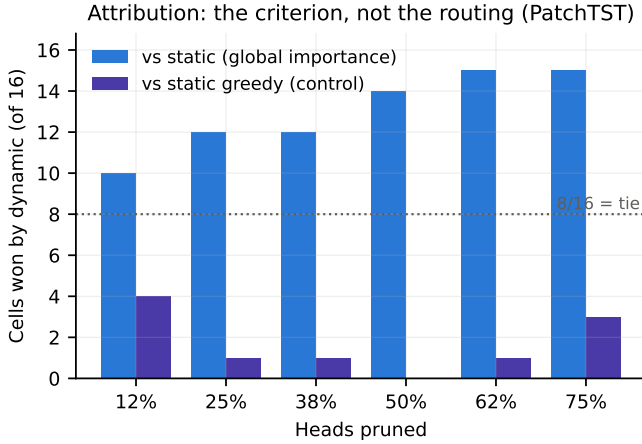


Fig. 4. Attribution across 16 dataset $\times$ horizon cells. Dynamic regime-routed selection wins up to 15/16 cells against the standard static baseline (blue), but only 0–4/16 against the pooled greedy control (violet): the selection criterion, not the regime conditioning, explains the gain.

heads attend to. The aleatoric-routing conclusion therefore applies to temporal-patch attention, with cross-variate attention marking its boundary.

G. Deployment notes

Per-window STL is the dominant serving cost of any regime-aware policy (194.5 ms/window on CPU); the learned detector reproduces its routing quality at 4.3 ms (45 $\times$ faster, 73–79% label accuracy, end-to-end MSE within noise of STL routing). Confidence-aware fallback to the static mask on ambiguous windows brings no additional gain at any threshold (Appendix Fig. 14). The practitioner recipe implied by our results: prune 25% of heads with pooled greedy selection and retrain with the mask (free); if accuracy is worth 2 $\times$ compute, average the top-2 specialist masks; do not build per-window routers on post-hoc masks.

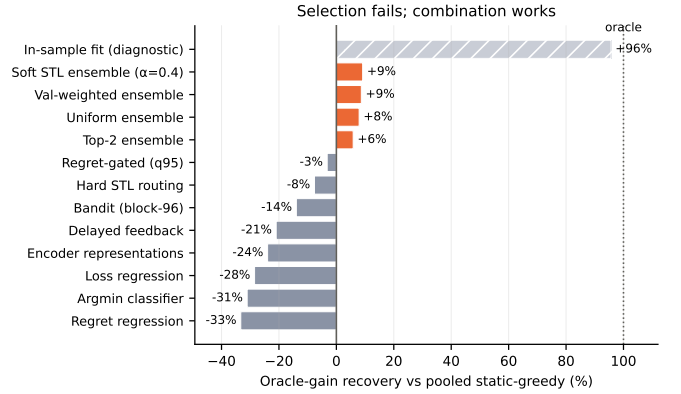


Fig. 5. Oracle-gain recovery relative to the pooled static-greedy mask. Every deployable router (grey) is net negative; prediction ensembles (orange) are the only positive deployable methods. The in-sample fit (hatched; not deployable) shows the router hypothesis class is expressive enough — the signal, not the model, is what is missing.

VI. CONCLUSION AND FUTURE WORK

We set out to test a natural hypothesis — that attention-head importance in time-series Transformers depends on the temporal regime of the input, and that regime-conditional head selection should therefore beat a single static mask — and subjected it to what we believe is the first criterion-controlled evaluation of dynamic pruning, over 16 dataset $\times$ horizon cells, 5 seeds and 6 sparsity levels.

The hypothesis survives in a sharper, redistributed form. Regime-dependent head structure is real within every trained model, and a per-window oracle over regime-specialist masks outperforms even the unpruned network almost everywhere. But none of that value is accessible through selection at inference time: once the selection criterion is held fixed, regime routing adds nothing on temporal-patch attention, and a hierarchy of twelve routers together with capacity,

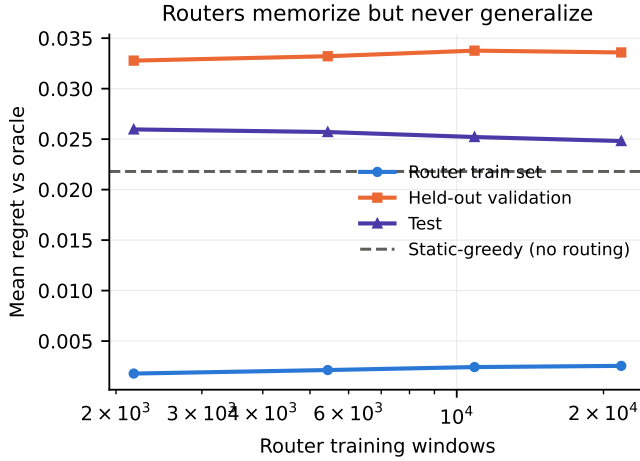


Fig. 6. Router learning curve (mean over @96 cells). In-sample regret stays near zero while held-out validation and test regret remain flat — and above the no-routing reference — as router training data grows 10 $\times$: routers memorize but never generalize.

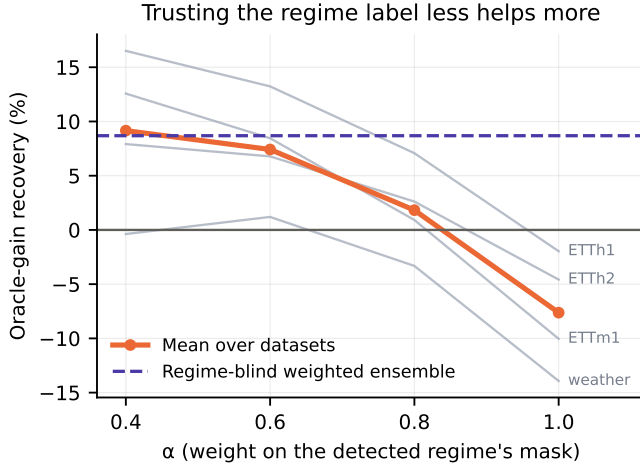


Fig. 7. Soft regime routing: the detected regime’s mask receives weight α , the others share $1-\alpha$. Oracle-gain recovery falls monotonically as trust in the regime label rises; the best soft variants only tie the regime-blind weighted ensemble (dashed).

data-scaling and distribution-shift diagnostics shows the per-window optimal mask to be aleatoric. What routing cannot capture, combination partially can: averaging the specialists’ predictions beats every router and the best static mask, and surpasses the unpruned model on two of four datasets — a diversity effect for which regime partitioning is the strongest, though not a necessary, source.

Alongside this analysis the study yields directly usable results: joint greedy selection replaces the fragile global-importance ranking at no extra conceptual cost; 25% of heads can be removed for free when the mask is imposed during training; a lightweight learned detector removes the 45 $\times$ STL bottleneck from any regime-aware deployment; and the conclusions transfer to iTransformer with a precise, mechanis-

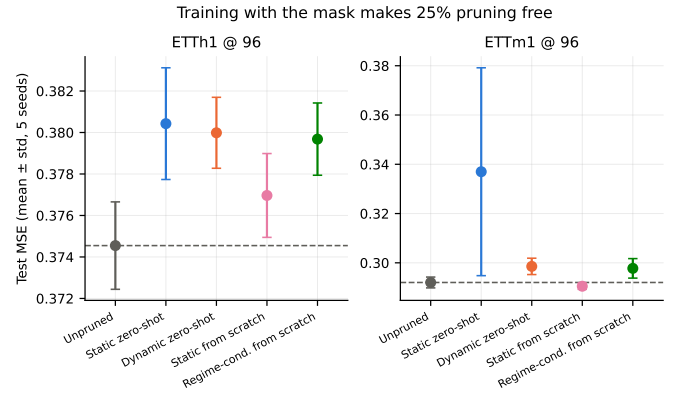


Fig. 8. Zero-shot masking versus masked-from-scratch training at 25% sparsity (test MSE, mean $\pm$ std over 5 seeds; dashed line = unpruned baseline). Training with the mask removes the zero-shot cost entirely; the regime-conditional training variant does not improve on plain masked training.

tically interpretable boundary — cross-variate attention retains a small genuine routing benefit that temporal attention does not.

More broadly, our results argue that claims about input-conditional sparsity should be required to pass two tests they rarely face: a selection-criterion-matched static control, and an oracle-normalized account of how much adaptive headroom exists and how much is actually recovered. We release per-window losses for every configuration to make such tests cheap for future work.

Our findings reframe rather than close the adaptive-sparsity question, and point to four directions for future work.

a) *Train the specialists to be routable.*: The masks we route between are found *post hoc* on a model never trained to be routable, so their per-window advantages are accidental — which is exactly what the aleatoric result reflects. Learning head gates end-to-end (a mixture-of-head-experts with a load-balancing objective, in the spirit of conditional-computation architectures [5, 9]) would make routing signal learnable by construction, and our oracle-normalized evaluation provides the yardstick for whether trained routability closes the gap that post-hoc routing cannot.

b) *Characterize the aleatoric residual.*: The oracle gap behaves like noise to every predictor we tried, but a formal account is missing: under what conditions on the mask-perturbation covariance and the target noise is per-window mask identity information-theoretically unpredictable, and how does the recoverable fraction scale with mask diversity? The released per-window losses make this a data-analysis problem rather than a training problem.

c) *Exploit the boundary.*: Cross-variate attention retained a genuine routing benefit where temporal attention did not. Testing whether this persists on strongly multivariate benchmarks (ECL, Traffic, larger Weather horizons), across capacity budgets, and for other inverted or hybrid architectures would turn a single boundary observation into a design principle for where adaptive head sparsity is worth building.

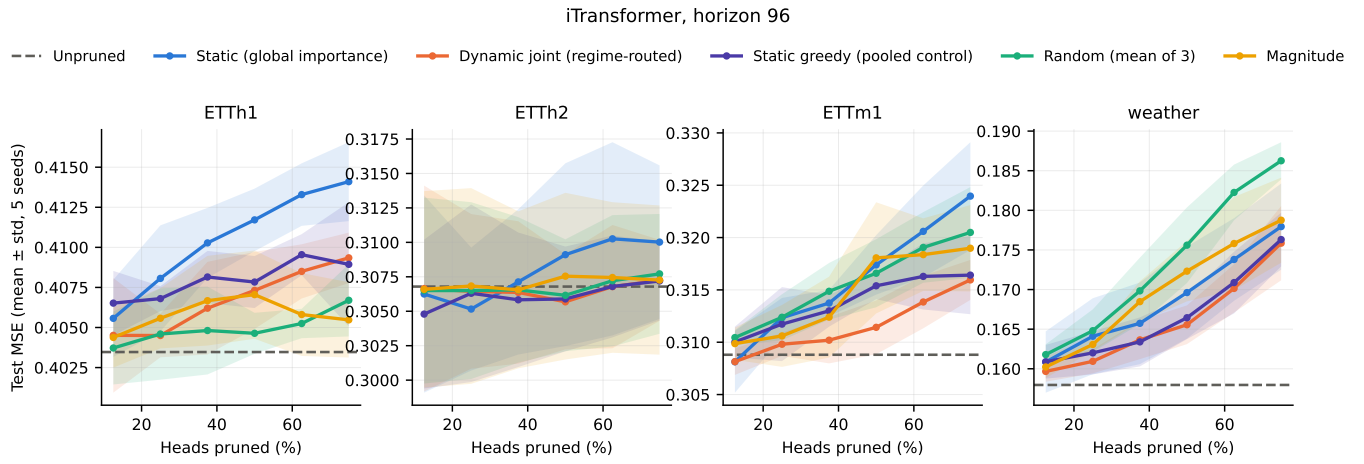


Fig. 9. The sparsity sweep on iTransformer (horizon 96). Cross-variate heads are far more redundant on the 7-variate ETT datasets — on ETTh2 pruning improves the model — while 21-variate Weather shows PatchTST-like degradation. The criterion finding replicates; a small regime-routing benefit survives the pooled control here, unlike on temporal-patch attention.

d) Cheaper and richer conditioning signals.: STL regimes are one coarse conditioning variable. Change-point detectors, learned soft regime embeddings, or exogenous co-variates may induce specialist pools with deeper *and more exploitable* structure; the diversity ablation suggests evaluating any such proposal on two axes — realized ensemble gain and oracle depth — rather than on routing accuracy alone. On the systems side, the top-2 ensemble’s $2\times$ forward cost can likely be reduced by sharing computation between masks that differ in only a few heads.

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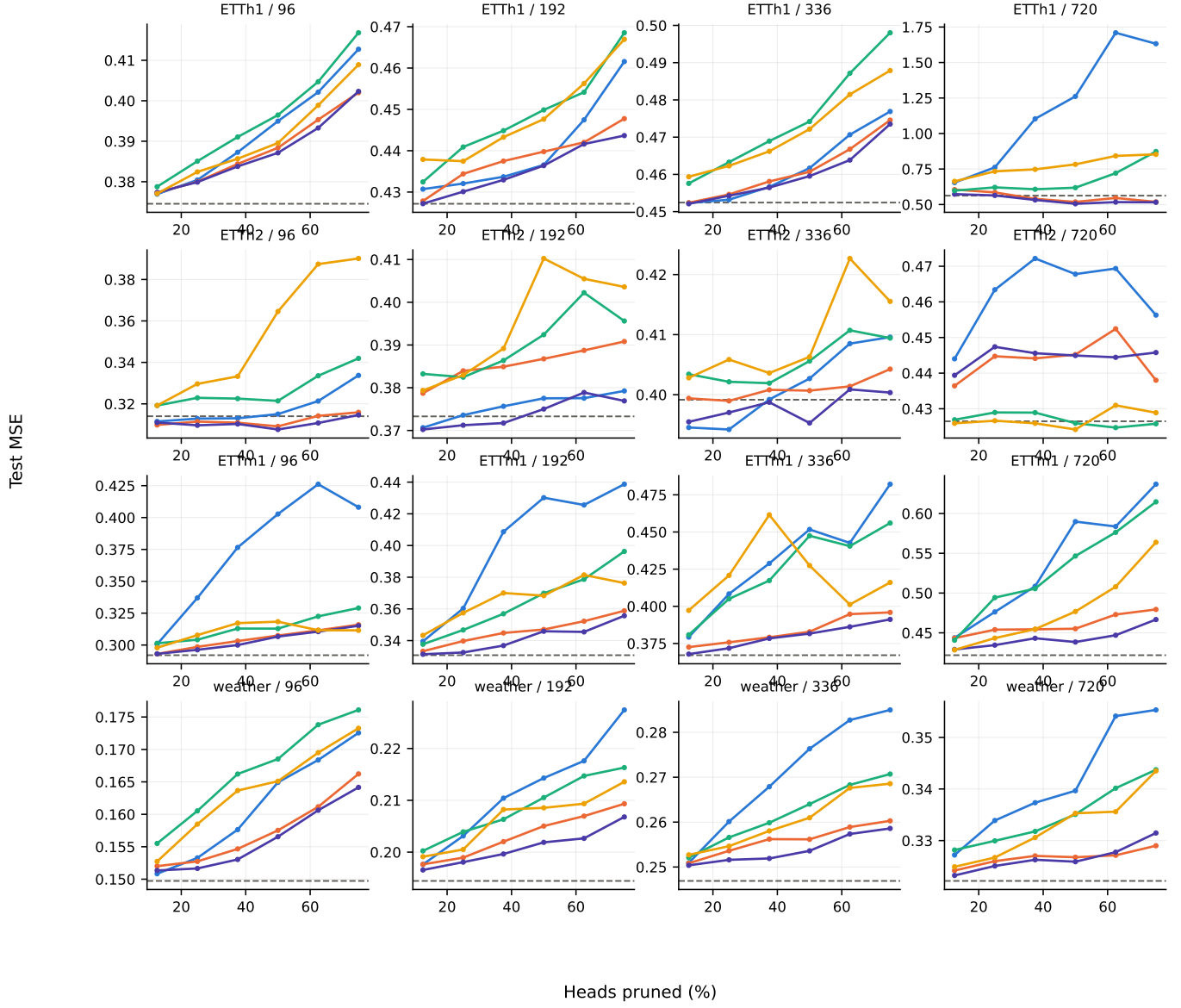


Fig. 10. The full grid: test MSE versus heads pruned for all 16 dataset×horizon cells (PatchTST; colors as in Fig. 3). Failures of the independent-importance ranking concentrate on ETTh1 and long horizons.

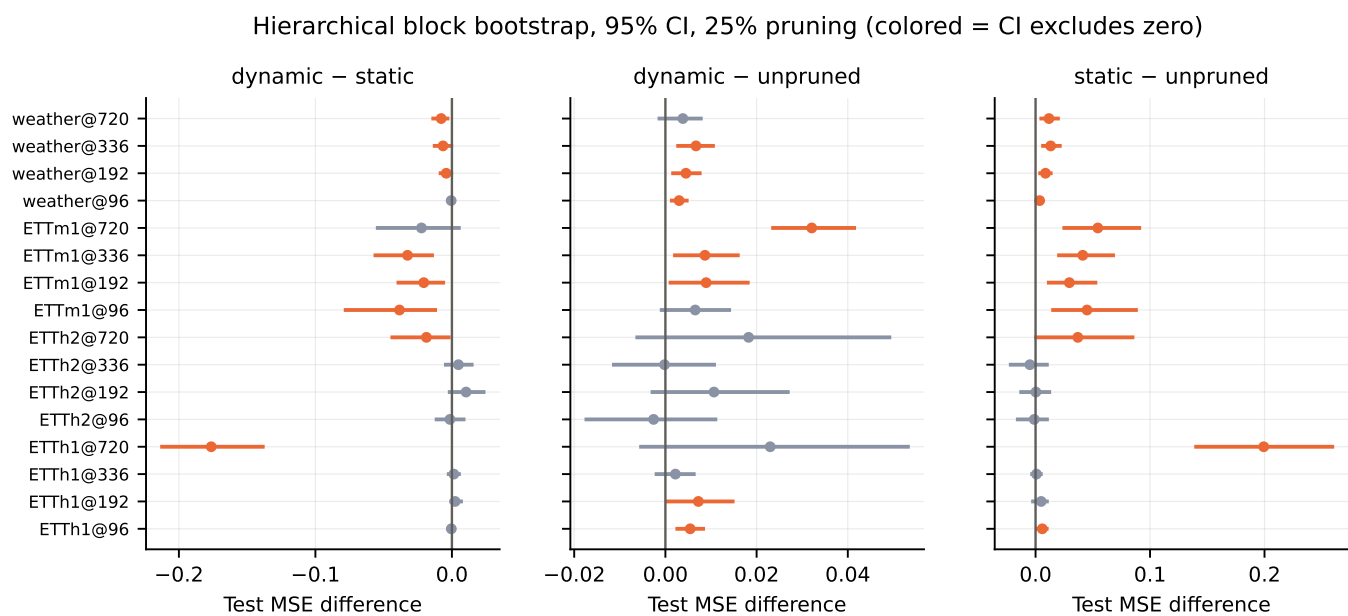


Fig. 11. Hierarchical circular moving-block bootstrap (seeds $\times$ 96-window blocks, 10^4 replicates) at 25% sparsity: 95% CIs for paired test-MSE differences per cell. Colored intervals exclude zero. Dynamic is significantly better than the standard static in 8/16 cells and never significantly worse (left); static is significantly worse than the unpruned baseline in 11/16 cells (right).

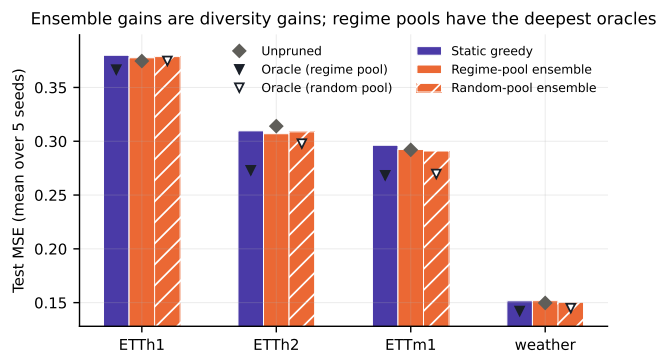


Fig. 12. Diversity-source ablation at horizon 96. Ensembles over regime-partitioned pools (solid orange) and random-subsample pools (hatched orange) perform comparably, but regime pools yield systematically deeper oracles (filled vs. open triangles) on all four datasets.

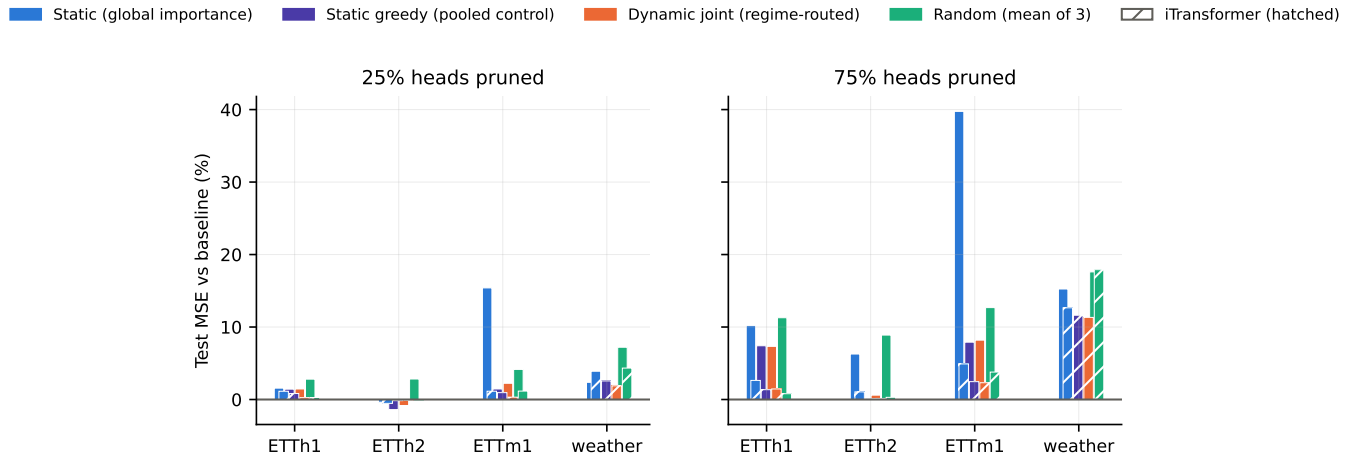


Fig. 13. Degradation at 25% and 75% pruning per method and dataset, PatchTST (solid) versus iTransformer (hatched), horizon 96. Cross-variate heads are substantially more redundant except on 21-variate Weather.

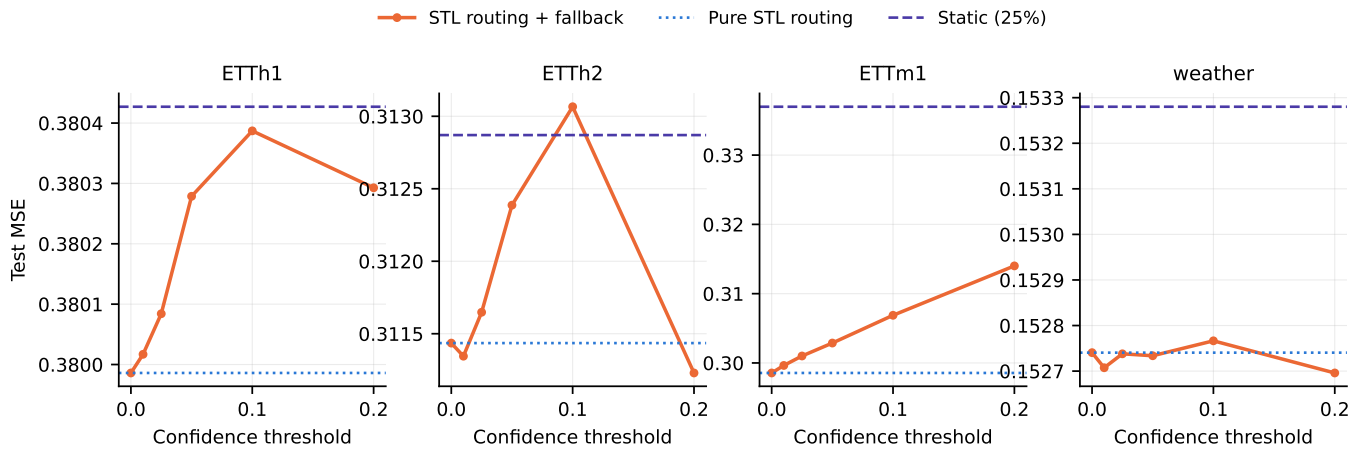


Fig. 14. Confidence-aware fallback at 25% sparsity, horizon 96: routing ambiguous windows (STL confidence margin below threshold) to the static mask never improves on pure STL routing, and inherits the static mask's weakness where it is fragile (ETTm1).